Grasping Force Feedback for da Vinci Surgical Robots

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***Abstract*—Robotic minimally invasive surgery lacks grasping force feedback, requiring surgeons to rely solely on visual cues during tissue manipulation. To address this limitation, we developed a kinesthetic haptic interface that provides grasping force feedback while maintaining the ergonomics of the da Vinci leader hand gripper. The device is a 1:1.2 scale replica actuated through a capstan-driven mechanism and controlled using a Hapkit 3.0 board with integrated force and touch sensing. A real-time Processing simulation synchronizes virtual tissue interaction with the rendered forces. Preliminary results show that the system can distinguish free-space, soft-tissue, and hard-tissue interactions, generating reaction forces between 0–4 N. Calibration of the force sensor produced a strong power-law fit (R^2 = 0.9723), supporting reliable force estimation. These findings demonstrate the feasibility of adding intuitive grasping force feedback to a da Vinci-style interface, with future work focused on hardware refinement, improved sensing, and user-study evaluation.**

I. Introduction

Minimally invasive surgery (MIS) has seen a significant shift toward robot-assisted procedures, driven by the promise of improved precision, reduced patient trauma, and faster recovery times. Among the commercially available platforms, the da Vinci surgical system (Intuitive Surgical) is the most widely adopted worldwide and has set the standard for clinical robotic surgery. Despite its proven advantages, the system lacks one critical capability: force (haptic) feedback during grasping. As a result, surgeons must rely exclusively on visual cues when manipulating tissues, often leading to excessive or inconsistent grasping forces. This deficit often leads to surgeons applying excessive or inconsistent grasping forces, which can damage delicate tissues. In [1], the authors show that providing grasping force feedback enables users to apply lower and more controlled forces and improves their intuitive understanding of tissue characteristics such as stiffness.

Several research efforts have attempted to address this limitation by integrating haptic feedback into surgical teleoperation. In [2], the authors introduce a pneumatic-based method that relocates finger forces to the wrist using custom, 3D-printed soft pneumatic actuators. Their experiments show that users apply significantly less force when assisted by haptic feedback. Another approach, presented in [3], utilizes a tendon-driven mechanism capable of displaying grasping forces in the 0-5 N range to the leader side, allowing for more realistic force cues during manipulation.

While these works demonstrate the value of grasping force feedback, most proposed solutions require substantial changes to surgeon interaction. They often rely on wearable pneumatic devices or redesign the surgeon-side master interface entirely. Such modifications disrupt the familiar ergonomics of clinically validated systems like the da Vinci and would require extensive retraining before adoption.

To address this gap, our team proposes a grasping force feedback solution that integrates directly with the existing da Vinci-style leader hand interface without altering its fundamental design. Our goal is to preserve the surgeon's established workflow while adding intuitive, physically grounded haptic cues. By maintaining compatibility with the da Vinci paradigm, the dominant platform in clinical MIS, our design moves toward a practical and adoptable enhancement for real-world surgical robotics.

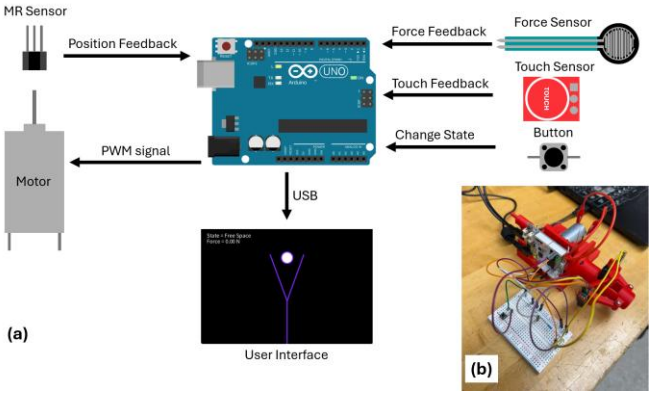


Fig. 1. (a) The Workflow of Grasping Force Feedback Mechanism; (b) Grasping Force Feedback Mechanism Prototype

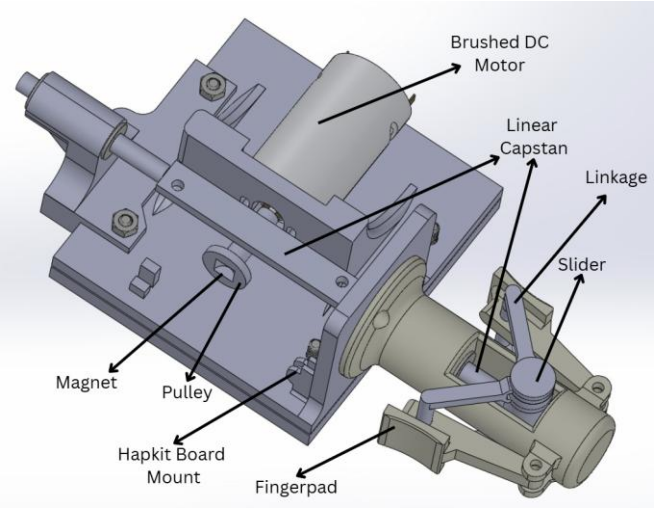


Fig. 2. Device CAD Model

II. METHODS

A. Device Description

The device is a kinesthetic haptic interface built as a 1:1.2 scale replica of the da Vinci 3 leader hand gripper. An overview of the system workflow, including the interaction between the hardware and simulation, is shown in Fig. 1. The entire mechanical assembly is 3D-printed using PLA and consists of a central linear slider connected to both finger pads through two symmetric linkage assemblies (see Fig. 2). The slider is driven by a capstan mechanism in which a linear rod attaches to a tensioned wire wrapped around a pulley actuated by a high-torque brushed DC motor. Sleeve bearings are incorporated along the linear capstan rod to minimize friction and ensure smooth, repeatable motion. By precisely controlling the motor position, the system can arrest the slider at designated points, enabling the device to render resistive forces to the user's fingertips.

The hardware is controlled using a Hapkit 3.0 board, which integrates a motor driver and a magnetic-resistive (MR) sensor for motor position encoding. A magnet mounted on the capstan pulley provides the reference for the MR sensor. A force sensor on the right finger pad measures user-applied reaction forces, and a touch sensor detects when the gripper is actively held; haptic feedback is enabled only when touch is detected. As a kinesthetic device, it primarily stimulates proprioceptive receptors such as muscle spindles and Golgi tendon organs, providing the user with a physical sense of resistance and tissue stiffness..

The main mechanical assembly, comprising the da Vinci gripper replica and linkage system, is mounted onto a base structure that houses the motor, capstan mechanism, and Hapkit board. The subassemblies are secured using M4 nuts and bolts.

A real-time simulation environment was implemented in Processing, as shown in Fig. 3. Position data from the Hapkit board are transmitted over USB to drive the virtual surgical gripper, which opens and closes synchronously with the physical device. When the virtual gripper contacts a simulated object, the software commands the motor to halt the slider accordingly, allowing the user to perceive the corresponding interaction forces through the device. Three states were displayed: free space, soft tissue and hard tissue

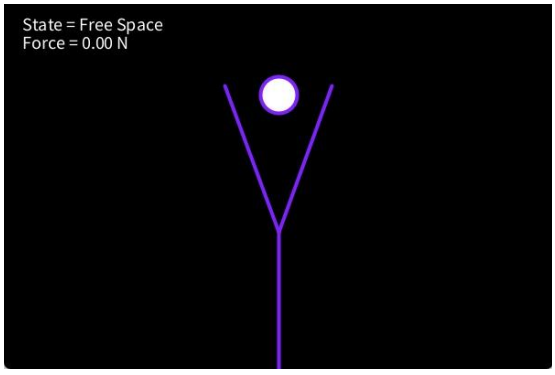


Fig. 3. Simulation in Processing

B. Preliminary Results

The force sensor resistor (FSR) was connected in a voltage divider configuration with a 10 kΩ fixed resistor, as shown in Fig. 4, allowing the sensor resistance to be computed from the measured output voltage using Eq. (1).

$$R_{fsr} = R_{fixed}\left(\frac{V_{cc}}{V_{out}} - 1\right) \quad \text{...Eq. (1)}$$

The resistance was then converted to force using the power-law relationship in Eq. (2),

$$F = aR_{fsr}{}^{-b} \quad \text{...Eq. (2)}$$

where the supply voltage was 5 V from the Arduino Uno. Calibration was performed using known masses to determine the relationship between force and sensor resistance. The result was well fitted by a power trendline, as shown in Fig. 5, giving constants a = 7961.8 and b = -1.034 with R^2 = 0.9723.

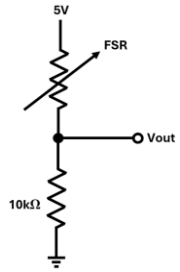


Fig. 4. Force sensing resistor voltage divider

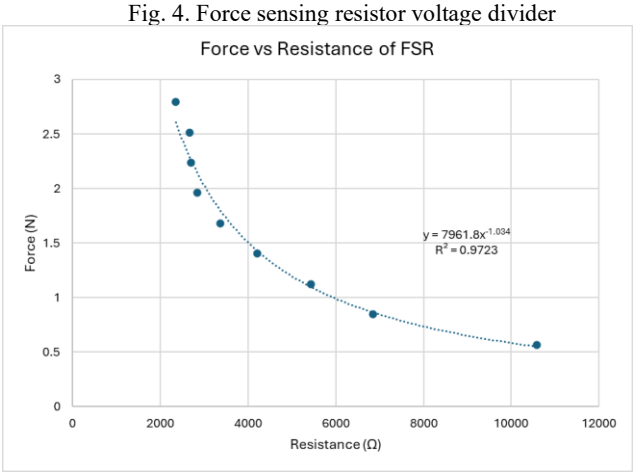


Fig. 5. Plot between force and resistance of FSR

Furthermore, the force was measured in the real system under different interaction states: free space, soft tissue, and hard tissue. The results are summarized in Table 1. Due to the nonlinear behavior of the system and time-varying finger position, the measured force fluctuated and could only be represented as a range rather than a single value.

In the free-space state, the measured force was approximately 0 N, with only internal friction present. The soft-tissue state produced higher forces in the range of 1–3 N, while the hard-tissue state resulted in the highest forces, ranging from 2–4 N. Notably, there was some overlap between the soft- and hard-tissue force ranges, reflecting variability in contact conditions and sensor nonlinearity.

Table. 1. Force in Different States

State	Stiffness (kN/m)	Force (N)
Free-Space	0	0-0.05
Soft-Tissue	200	1-3
Hard-Tissue	600	2-4

III. FUTURE WORK

A. Future Development

Future improvements to the device focus on both hardware and software enhancements. On the hardware side, upgrading the current brushed DC motor to a brushless DC motor would enable smoother, more precise force rendering and reduce mechanical noise. The magnetic-resistive (MR) sensor, while functional, requires periodic recalibration and is prone to position drift; replacing it with a higher-resolution and more robust encoder would improve reliability. Mechanically, the linkage and slider assembly should be redesigned to reduce friction and eliminate singular configurations that limit controllability. Optimizing the slider travel and linkage geometry would yield a more stable and repeatable haptic response.

On the software side, a more sophisticated simulation environment would significantly improve user perception and training value. Replacing the current Processing-based stick-figure visualization with a higher-fidelity physics and graphics engine would allow more realistic object interaction, clearer representation of tissue deformation, and better alignment between the rendered forces and the displayed virtual environment.

B. User Study Protocol

A user study will be conducted to evaluate the grasping force feedback performance of the haptic device. The study is designed as a psychophysics experiment to measure the just noticeable difference (JND) in perceived tissue stiffness. The experimental protocol will be based on the method described in [4]. Sixty healthy participants aged 20–45 will be recruited. All participants will have no prior experience with handheld instruments or minimally invasive surgical robots and no known tactile or motor impairments. A 10-minute training session will be provided before data collection to familiarize participants with the device and reduce learning effects.

During the experiment, participants will interact with the haptic device to distinguish between simulated soft and hard tissue conditions using grasping motions. In each trial, participants will report whether the presented stimulus felt closer to soft or hard tissue. The one-up two-down algorithm will be used to vary the force feedback level based on participant responses, allowing convergence toward the JND threshold. The minimum force difference at which participants can reliably discriminate between soft and hard tissue will be recorded as the primary performance metric.

C. Evaluation Metrics

The performance of the haptic device will be evaluated using both quantitative and qualitative metrics. The primary quantitative metric is force discrimination threshold, measured in terms of the just noticeable difference (JND). This metric represents the smallest force difference that participants can reliably perceive when distinguishing between simulated tissue stiffness levels. In addition, response time will be recorded to measure how quickly participants can identify differences in force feedback. Response time is an important metric in surgical contexts, where delayed perception of force may increase task difficulty and risk.

Qualitative evaluation will focus on perceived feedback intuitiveness, which reflects how naturally and easily participants interpret the force feedback provided by the device. This metric will be assessed using post-experiment questionnaires with Likert-scale ratings and open-ended comments. Collecting qualitative feedback will provide insights into the usability and perceived effectiveness of the haptic design, complementing the quantitative performance measurements.

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